



LibreOffice GSoC Contributor Proposal Template

Summary : This project aims to add support for importing JPEG XL (.jxl) images into LibreOffice. By integrating a modern image decoding library and expanding LibreOffice's existing image filtering system, the project will enable users to seamlessly insert and display JPEG XL images in Writer, Impress, and Draw, improving compatibility with next-generation image formats..

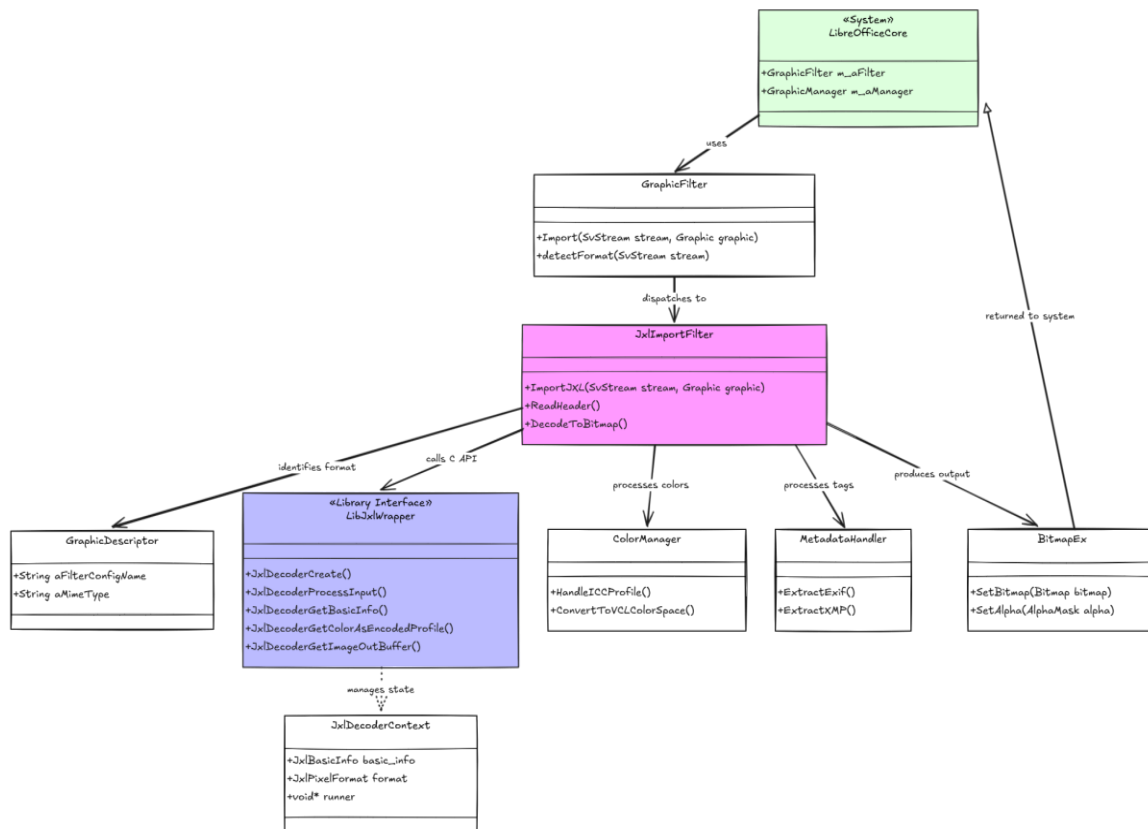
Project Abstract

LibreOffice currently lacks support for the JPEG XL image format (tdf#156931), which results in .jxl files being treated as attachments rather than images. JPEG XL is a modern image format designed to provide superior compression, high image quality, and advanced features such as lossless compression, progressive decoding, HDR support, and animation.

This project aims to implement JPEG XL import functionality in LibreOffice using the **libjxl** library. The implementation will follow the design patterns used in previous image format integrations such as WebP. The goal is to allow users to open and insert JPEG XL images naturally within documents while maintaining performance and compatibility with LibreOffice's graphics system.

The deliverables include implementing the decoding pipeline, integrating it into LibreOffice's filter system, validating correctness, and ensuring proper handling of color spaces and image metadata.

Illustrative diagram reflecting my understanding of the **Project Abstract**, for demonstration purposes only:



Background

LibreOffice supports many common image formats, but lacks support for newer formats like JPEG XL.

This leads to several limitations:

- Missing format support: .jxl images cannot be inserted directly as images.
- Compatibility issues: Users must convert images to older formats before use.
- Growing adoption: JPEG XL is increasingly supported by modern tools and platforms.

JPEG XL offers significant improvements over traditional formats:

- Better compression efficiency with high image quality.
- Support for both lossy and lossless encoding.
- Advanced features like progressive decoding, HDR, and animation.

Adding support for JPEG XL will improve LibreOffice's compatibility with modern workflows and enhance user experience.

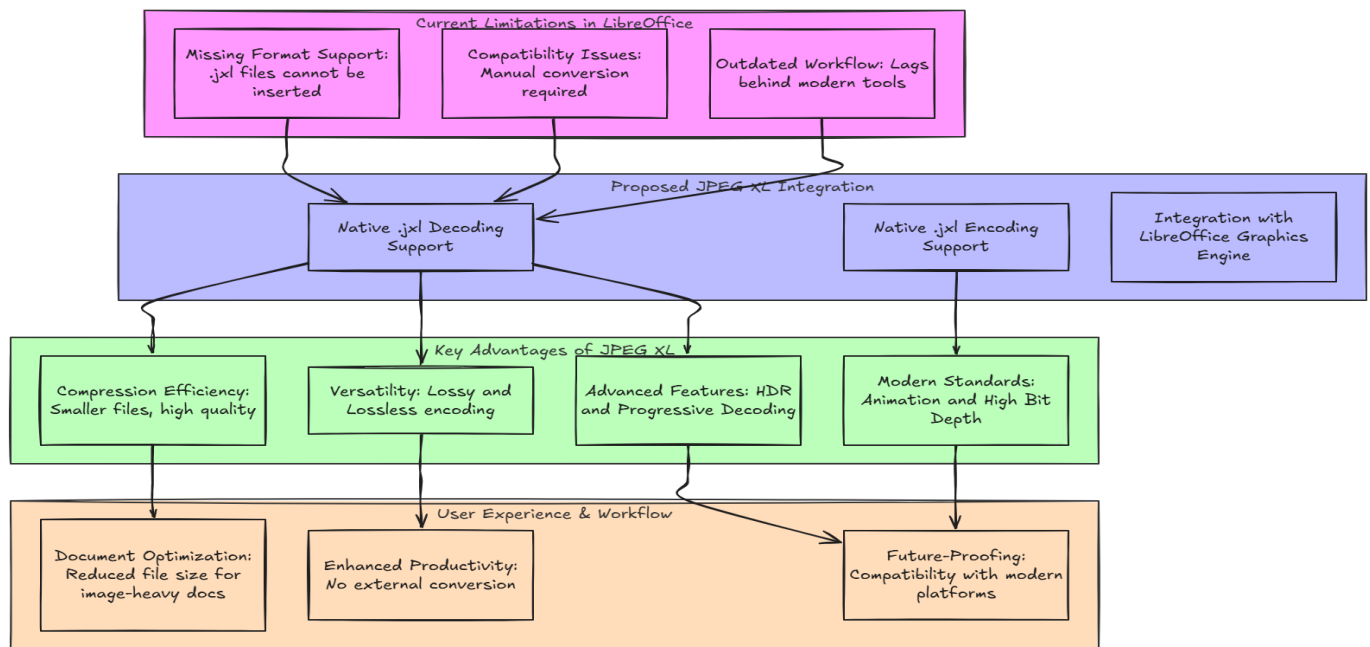
References

- The Document Foundation GSoC Ideas Page – JPEG XL Support
- LibreOffice Bug Tracker – tdf#156931
- libjxl (Reference implementation and decoder library)
- ISO/IEC 18181 – JPEG XL Image Coding System
- LibreOffice source code (WebP implementation as reference – `vcl/source/filter/webp/`)

References

- The Document Foundation GSoC Ideas Page – JPEG XL Support <https://wiki.documentfoundation.org/Development/GSoC/Ideas>
- LibreOffice Bug Tracker – tdf#156931 https://bugs.documentfoundation.org/show_bug.cgi?id=156931
- libjxl (Reference implementation and decoder library) <https://github.com/libjxl/libjxl>
- ISO/IEC 18181 – JPEG XL Image Coding System <https://jpeg.org/jpegxl/>
- LibreOffice source code (WebP implementation as reference) <https://github.com/LibreOffice/core/tree/master/vcl/source/filter/webp>
-

Illustrative diagram reflecting my understanding of the Background, for demonstration purposes only:



Design Ideas / Approach

Design Ideas / Approach

- The project will implement JPEG XL import support through the following steps:
 1. **Study Existing Implementation:**
Analyze WebP import support in LibreOffice to understand filter integration.
 2. **Library Integration:**
Integrate libjxl into LibreOffice's build system and ensure proper linking.
 3. **Decoder Implementation:**
Use libjxl APIs to decode .jxl images into a format compatible with LibreOffice's graphics engine.

4. Filter Integration:

Extend LibreOffice's image filter framework to recognize and load JPEG XL images.

5. Testing & Validation:

Test with various .jxl images (different sizes, color spaces, alpha channels).

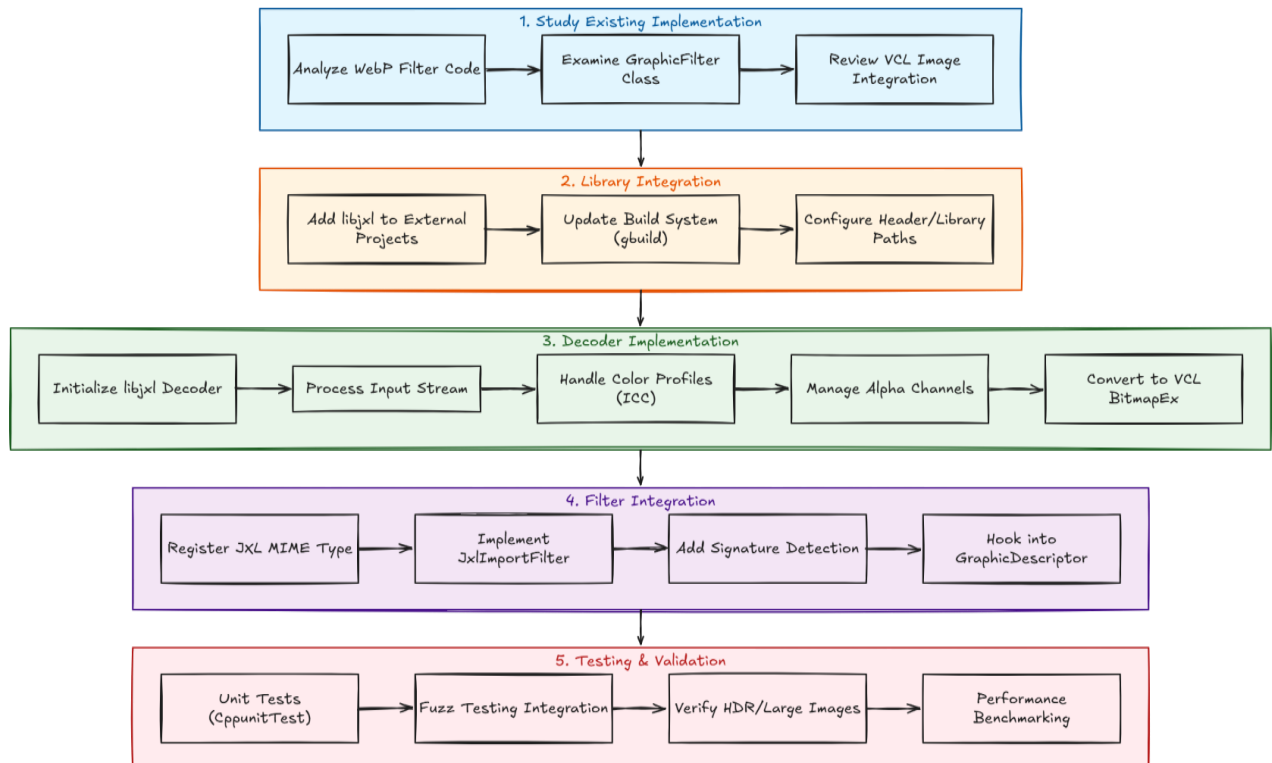
6. Performance Considerations:

Ensure efficient memory usage and acceptable decoding speed.

Tools & Technologies:

- C++
- LibreOffice codebase
- libjxl (JPEG XL reference implementation) • Git & Gerrit

Illustrative diagram reflecting my understanding of the Design Ideas , for demonstration purposes only:



Code Affected

- The exact components will be identified after initial exploration of the LibreOffice codebase.
- The implementation will mainly focus on:
- Image filter and import system in LibreOffice
- Integration of libjxl decoder into graphics pipeline
- Handling image decoding and conversion to internal bitmap format
- Supporting color profiles, alpha channels, and metadata

Project Goal / Impact

- The main goal of this project is to **add native JPEG XL image import support into LibreOffice**, enabling users to seamlessly open and insert `.jxl` images just like any other supported format.
- This contribution will have a meaningful impact on both users and the open-source ecosystem:

For Users

- Enable direct insertion of modern image formats without conversion.
- Improve workflow efficiency and reduce friction.
- Enhance visual quality in documents through better compression and HDR support.

For LibreOffice

- Keep the platform aligned with modern standards in image processing.
- Strengthen LibreOffice's position as a competitive open-source office suite.
- Expand compatibility with external tools and ecosystems adopting JPEG XL.

For the Community

- Contribute to the adoption of an open, royalty-free image standard.
- Provide a reusable implementation that can be extended to support export in the future.

Related Work / Pre proposal Work

I reviewed LibreOffice Bug tdf#55960 (https://bugs.documentfoundation.org/show_bug.cgi?id=55960), or direction link to solve : <https://bug-attachments.documentfoundation.org/attachment.cgi?id=206477>) and its attached patch (attachment 206476) to understand how the fix addresses floating-point canonicalization in FillSeries.

I also studied the JPEG XL format, its compression techniques, decoding structure, and advantages over traditional formats. Additionally, I explored how other image formats like WebP are integrated into LibreOffice to understand the expected architecture and workflow.

Previously, I developed a buffer pool manager (https://github.com/Moemenmohamed24/Buffer_Pool_Manager), where I worked on memory management, caching strategies, and performance optimization using algorithms like LRU-K. This experience strengthens my ability to handle performance-critical decoding and memory usage efficiently, directly relevant for this project.

Schedule of Deliverables (timeline)

May 1 – May 26 (Community Bonding Period)

May 1 – May 12:

- Connect with mentor and discuss project scope
- Study LibreOffice architecture and image filter system
- Set up development environment and build LibreOffice

May 13 – May 20:

- Analyze WebP import implementation as reference
- Study libjxl API and decoding workflow
- Run initial experiments with libjxl decoding

May 21 – May 26:

- Plan implementation details for Phase I

May 27 – July 12 (Phase I)

May 27 – June 2:

- Integrate libjxl into build system
- Prepare basic decoding prototype

June 3 – June 9:

- Implement JPEG XL decoding into bitmap format
- Begin integrating into LibreOffice filter system

June 10 – June 16:

- Complete basic import functionality
- Test with simple .jxl images
-

June 17 – June 23:

- Improve handling of color profiles and alpha channels
- Fix bugs and refine decoding

June 24 – June 30:

- Add support for more complex images (large size, HDR)
- Write unit tests

July 1 – July 7:

- Code cleanup and optimization
- Prepare documentation

July 8 – July 12:

- Submit Phase I for evaluation

July 13 – August 19 (Phase II)

- Improve performance and memory handling
- Extend support (multi-frame / advanced features if feasible)
- Address mentor feedback and finalize implementation

August 20 – November 11 (Extended Timeline, if needed)

- Additional refinements and optimizations
- Ensure stability across platforms

Note: I have final exams in May and June, which may affect my availability for full-time work during these months. I will manage my schedule to balance study and project commitments.

Communications

Timezone / Working hours:

Egypt Time (UTC+2), 15–20 hours/week

I am highly committed to my work and always respect deadlines. I maintain continuous communication, provide regular updates, and actively seek feedback. I am flexible and available for discussions at any time using any platform preferred by the mentor or organization.

Contact:

- Email: moamenhamous@gmail.com
- GitHub: <https://github.com/Moemenmohamed24>
- Weekly check-ins via email and GitHub issues
- Available on WhatsApp, Discord, Zoom, Google Meet, or any preferred platform
- LinkedIn: [linkedin.com/in/moemen-mahmoud-337a6628b](https://www.linkedin.com/in/moemen-mahmoud-337a6628b)

About Me

I am Moemen Mahmoud Hamous, a computer engineering student.

I have experience in C++ , C# and system programming, along with strong knowledge of algorithms and performance optimization.

I have worked on projects such as buffer pool managers, banking systems, and problem-solving applications. I am familiar with Git, GitHub, and software design principles (OOP & SOLID).

I enjoy writing clean, efficient code and contributing to open-source projects.

Prior Experience with open source

Contributed to public repositories on GitHub, mainly exploring "good first issues." Completed real-world R&D volunteer work (Grido Organization) and participated in team-based software simulations (Aspire Consulting).

Why LibreOffice?

LibreOffice is one of the most widely used open-source office suites, impacting millions of users worldwide. Contributing to it means improving a real-world application used daily for documents, presentations, and data processing.

I am particularly excited about this project because it involves adding support for a modern and future-proof image format like JPEG XL. This directly improves user experience and keeps LibreOffice up to date with evolving technologies.

What motivates me is the opportunity to work on a real feature requested by users (tdf#156931) and contribute something meaningful to the community. I am also excited about collaborating with experienced mentors and learning from the LibreOffice development ecosystem.

This project aligns perfectly with my interest in C++, systems programming, and performance-aware development. I am eager to contribute high-quality work, learn from the process, and deliver a solid implementation that benefits LibreOffice users.